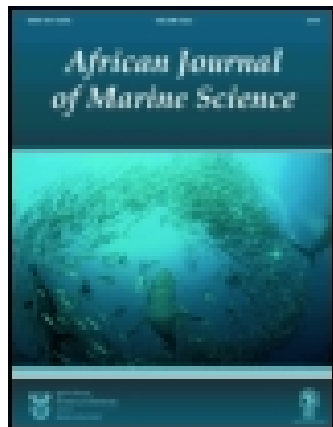




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Sustainability and present-day approaches to fisheries management — are the two concepts irreconcilable?

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Sustainability may be defined as the maintenance of the quality, diversity and availability of fishery resources in sufficient quantities for present and future generations. But how do modern management systems aim to achieve this in the face of natural fluctuations and human-induced pressures? Using four case study fisheries (northern Atlantic cod and North Sea cod [both *Gadus morhua*], Pacific sablefish *Anoplopoma fimbria*, and South African West Coast lobster *Jasus lalandii*), we examine the components and considerations used in the development of management systems. Historically, these considerations have been biology-based, incorporating reference points appropriate for management goals or targets. Following some quite spectacular failures, however, inputs from disciplines such as socio-economics and ecosystem science are increasingly important drivers behind management

decisions, but have the potential to increase management complexity greatly. We identify three areas of particular importance. Precautionary, robust and responsive management is needed, with pre-agreed actions to be taken when particular conditions are experienced. Management decisions need to be based upon adequate scientific information, particularly given the increasing variety of drivers for management. Finally, fisheries management needs to be a collaborative process, involving all stakeholders, thereby raising the credibility of management for all participants. Integrating all these aspects within the fisheries management framework is complex, but when combined with the realisation that harvests are likely to be lower than historical levels, there appears to be a greater chance of successful sustainable exploitation.

Keywords: fisheries management, precautionary approach, sustainability

Introduction

Like many scientific and advisory disciplines, fisheries management is a simple concept to introduce, but a complex one to embrace. Since fishing developed beyond being a subsistence or artisanal activity, the complexity of the scientific and advisory process for fisheries management has increased significantly (Beverton 1998). Traditional stock assessment approaches that underpin fisheries management developed rapidly during the 20th century, but the seminal work of Beverton and Holt (1957) marked a step-up in the numerate process by which fish stocks are evaluated. There have been many subsequent advances, particularly as computing power increased, but they all seek the same outcome, namely sustainability or sustained output from resources that are, by their very nature, highly dynamic. In this sense, sustainability may be defined as the maintenance of the quality, diversity and availability of fishery resources in sufficient quantities for present and future generations. The dilemma is how modern management systems can achieve this in the face of natural fluctuations,

human-induced changes in the resource and ecosystem, inadequate enforcement, and changing political drivers.

Early practitioners probably believed that successful fishery management required only a good understanding of the biology of fish and shellfish stocks (i.e. their biology and dynamics). The fact that many stocks are now in trouble and require rebuilding (FAO 2000, Caddy and Agnew 2004), however, shows that fisheries science and management may well be failing to put the concept of sustainability into practice. It is now broadly accepted that many other disciplines interact with natural science, so, for instance, there have been spectacular breakdowns in fisheries management systems where managers and decision-makers have seemingly not understood the complex interrelationships between economics, sociology and natural science that exist in fishing and fisher behaviour (Branch *et al.* 2006).

Recognising that all was potentially not well with fisheries management, although not providing the reasons why, the UN's FAO Committee on Fisheries (COFI) in 1991 called for

the development of new concepts to lead to responsible, sustained fisheries, which in 1995 culminated in the production of a Code of Conduct for Responsible Fisheries (FAO 1995). During the development of that code, Garcia (1994) advocated the use of a number of 'tools' to be used by managers and decision-makers when applying the precautionary principle in fisheries management:

- Adopt the sustainable development principle
- Adopt the principle of precautionary management
- Use the best scientific evidence available
- Adopt a broad range of management benchmarks and reference points
- Develop criteria for use when assessing the impacts of development
- Take a risk-averse stand
- Agree acceptable levels of impacts and risk
- Take a holistic view of resources within their environment
- Speed up management response time
- Allow for greater participation by non-fishery users in management bodies
- Improve decision-making procedures
- Introduce prior consulting procedures
- Strengthen monitoring, control and surveillance (MCS)

Use of the term 'sustainability' is noteworthy, as are ideals such as 'risk-aversion', 'holistic view' (= an ecosystem approach), and 'reference points'. Considering these issues has made present-day fisheries management more complex, and all are now receiving crucial research and administrative attention. For instance, although there are not yet clear examples of the implementation of an ecosystem approach to fisheries management, there is huge investment in trying to drive management in that direction. Notwithstanding, the word 'precautionary' remains in fisheries management because managers and decision-makers realise that there is no simple fix when balancing the complexity of the conflicting issues that need to be considered. This then leads to the question in our title: are sustainability and present-day approaches to fisheries management irreconcilable?

Case Studies

Here we look at four case studies from around the world (Figure 1) and ask the question whether they really are achieving their primary aim of sustainability. We take most of the information on each from the literature and in certain cases from personal knowledge. We also investigate whether precautionary management, as defined by Garcia (1994) as underpinning sound fisheries management, is actually taking place. Readers interested in the detail of these case studies are referred to the literature sources cited; we aim here only to overview the extensive literature on each resource. For each case we mention a few of the issues that could be confounding the attainment of sustainable fisheries management.

North Sea cod

Cod *Gadus morhua* in the North Sea was once the mainstay of the UK fishing industry. Currently, however, the

stock is heavily exploited and seriously depleted, suffering from reduced recruitment, and showing little sign of sustained recovery (Cook *et al.* 1997, Bannister 2004, Horwood *et al.* 2006). Fisheries management, as assessed against the benchmark of sustainability, is surely therefore failing. Although fishing mortality has been reduced, real recovery is hampered by considerable uncertainties in the stock assessments as a consequence of, *inter alia*, low stock size, variability in the fisheries-independent survey indices, and inaccurate catch data (Horwood *et al.* 2006). The reduction in fishing mortality to its current level, well below what it was in the heyday of the cod fishery, has simply been inadequate to assure recovery of the stock in a reasonable amount of time, especially given the mixed species nature of the gadoid fishery (which targets whiting and haddock as well as cod) and the dynamics of the environment in which the stock exists (Kell *et al.* 2005b).

There is no doubt that the North Sea cod fishery has been impacted by increasing fishing power, but it is with the issues of likely inaccurate reporting, plus delays in bringing stronger management measures to bear, where the problem probably lies (Bannister 2004). Not all countries neighbouring and fishing in the North Sea are part of the EU system, and some countries within the EU may have different objectives in terms of fishing needs. This overexploitation of a single resource by a number of countries represents the 'Tragedy of the Commons' if not at a national level, then at a regional level.

In any event, despite massive investment in research on cod in the North Sea, as well as direct and intuitive attempts to bring to bear on the fishery the precautionary management tools of Garcia (1994), the result has not been very successful. In our opinion, it is questionable indeed whether the sustainable development principle has been invoked for the stock, although anticipated precautionary management tools have been used over the years.

Where North Sea cod has done well in terms of management is in at least taking cognisance of the best scientific advice available and in looking at the stock and its environment/ecosystem holistically. Massive databanks of scientific information have been collected on the latter issue. Monitoring and surveillance have been good, concomitant with the stock's historical importance and its accessibility to many European countries, with Vessel Monitoring Systems (VMSs), logbooks, at-sea inspections, and (limited coverage) observer programmes. In turn, a range of management benchmarks and reference points has been adopted for North Sea cod, and scientists have developed rigorous criteria against which to assess the impact of development, as well as agreeing acceptable levels of impact and risk. However, it seems that unanimity of opinion on those levels between all stakeholders has yet to be reached. Despite the annual cycle of assessment and management, the management response time for the North Sea cod fishery, as for many fisheries, tends not to be speedy enough to preclude lasting damage being done when problems arise, and it is only somewhat recently that non-fishery users have been truly involved in management, through the EU-instituted and very active North Sea Regional Advisory Council (RAC). However, historical management of North Sea cod cannot

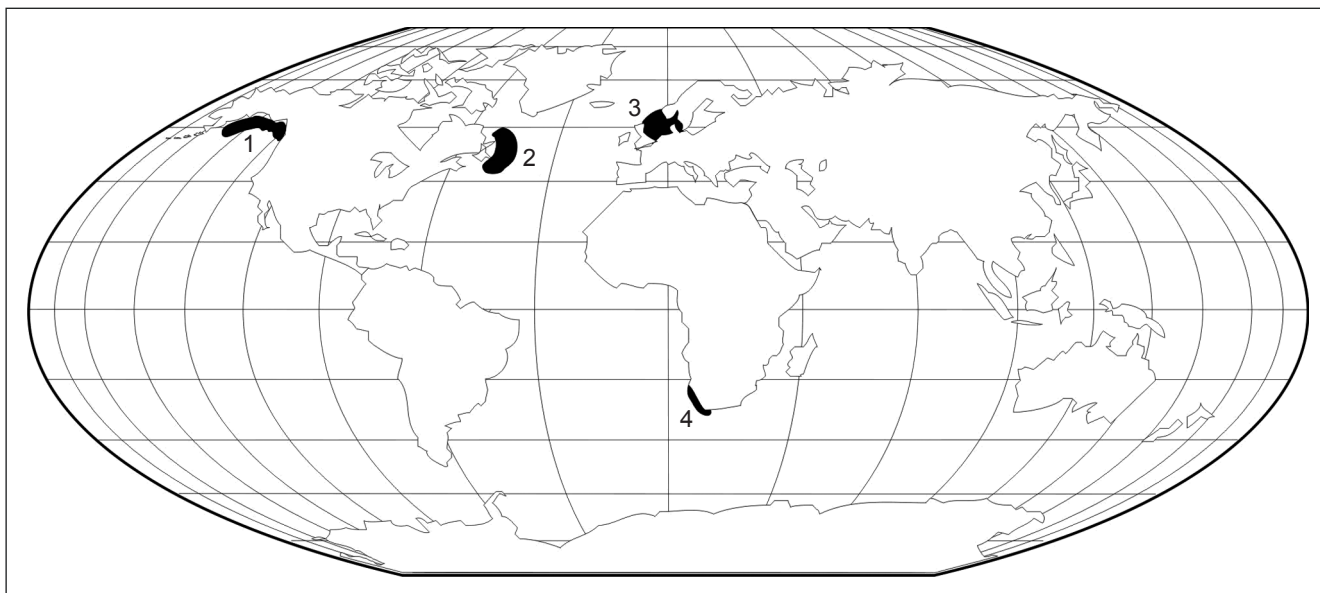


Figure 1: World map showing the location of the four case study fisheries discussed here: 1 Sablefish; 2 Northern Atlantic cod; 3 North Sea cod; 4 South African West Coast rock lobster

be considered to have been even mildly risk-averse, and along with that, decision-making procedures have hardly changed while the stock status was clearly declining. In turn, the recovery plan adopted by the EU Council in 2004 (Council Regulation (EC) 423/2004) is still to be fully implemented. Recovery is required before the long-term management plan agreed between the EU and Norway in 2005, which offers the potential for improved pre-agreed decision-making, can be implemented.

In the case of the North Sea cod stock, therefore, even state-of-the-art scientific evidence is probably insufficient to save it from at least commercial extinction, as Horwood *et al.* (2006) attest, although it may be among the most researched stocks anywhere in the world. Stakeholder involvement in management of North Sea cod is increasing at present, and extra data on the stock are being collected through a government-funded science/industry partnership (Armstrong *et al.* 2008). These improvements are all good for management of cod, but externalities such as the changing environment in the North Sea (O'Brien *et al.* 2000, Horwood *et al.* 2006) still jeopardise the chances of recovering the stock to historical levels of biomass. Consequently, therefore, there is no certainty that North Sea cod can be managed and maintained even at current levels of biomass, and recovery to historical levels may not be feasible.

Northern Atlantic cod

The northern Atlantic cod (also *Gadus morhua*) stock is located off the Newfoundland and Labrador coasts. It represents perhaps the most quoted occurrence of a collapsed stock, and the problems for fisheries science and management in ensuring sustainable exploitation of a commercial fish stock (Shelton 2005). Prior to its collapse, the stock averaged almost 50 000 t annually and had been exploited since

the 1400s (Taggart *et al.* 1994). However, despite increased controls and surveillance following the declaration of the Canadian 200 nautical mile Exclusive Fishery Zone in 1977, which led to a short recovery, stock size declined rapidly in the late 1980s and early 1990s, and a moratorium on fishing was put in place in July 1992. The expected recovery of the stock following the cessation of fishing has not materialised.

International annual assessments of the cod stock are performed under the auspices of the Northwest Atlantic Fisheries Organization (NAFO) and within national laboratories. NAFO aims to 'achieve the optimum utilization of the fishery resources in the Regulatory area' (Article XI of the NAFO Convention), examining proposed actions within the Regulatory area and by Coastal states, taking into account potential species linkages, thereby defining a criteria to assess the impact of development. However, involvement or consultation of fishery users is not immediately apparent within the process, although the latter is likely to have taken place through direct or indirect informal routes. The assessments performed use both annually updated commercial and fishery-independent survey data, and requests have been made to increase the rate at which data are provided to facilitate assessments (NAFO 1992). A number of state-of-the-art assessments are applied, to ensure that the best scientific evidence available is used. These are compared with appropriate stock reference point values, and results inform management decisions on allowable catches. However, no explicit agreement of acceptable risk levels is defined, despite the use of those precautionary reference points.

Scientific concern was frequently raised on the status of the stock (e.g. Harris 1990). Precautionary total allowable catch (TAC) levels were set, but the inability of management to halt cod stock declines (suggesting insufficient precautionary and timely management action), and the low

level of the stock that had been reached meant that recovery was reliant on the arrival of strong year classes, which did not appear.

The decline in northern Atlantic cod provoked a number of investigations into how it happened, and in particular increased published study of why there has been no recovery. Many of these build upon studies ongoing prior to the collapse, suggesting that the holistic view of resource management was developing. Despite the increased scientific study following collapse, however, there is still no widespread agreement on either. Indeed, more than 40 different hypotheses have been examined (Rice *et al.* 2003).

Fishing impacts alone have been suggested as sufficient to have caused the collapse of the stock, combined with high levels of discarding that led to significant mortality of immature cod (Myers *et al.* 1996). Assessments overestimated stock size during the 1980s (Sinclair *et al.* 1991), and biases in the assessment of biomass and estimates of catch per unit effort, attributable to improving fishing techniques, discarding and contraction of stock ranges with decreasing stock size (Hilborn and Walters 1992, Rose and Kulka 1999), increased these problems (but see Shelton 2005). The overestimation of stock size was recognised and the assessments revised downwards as a result, but despite this, the extent of the decline was underestimated. Fault cannot be attributed to the assessment alone, however. Shelton (2005) suggested that, even if a pessimistic assessment had been presented in 1989, there was strong resistance then to reducing TACs by the extent required by the scientific assessments. In turn, more flexible and reactive management, allowing within-year revision of quota, would have allowed new information to be taken into account (Caddy and Agnew 2004).

Environmental variability may also have played a role in the collapse of this stock. Changes in the environment during the decline may have shifted cod distribution and decreased individual condition and overall recruitment (Rose *et al.* 2000). In addition, there were increases in the population of pelagic fish and grey seals, which may have increased the natural mortality of cod at a variety of life-history stages (Swain and Sinclair 2000). However, increases in natural mortality are unlikely to be solely responsible for the stock collapse, and fishing mortality is felt likely to be the primary cause (Fu *et al.* 2001).

Further studies have suggested that the failure of northern Atlantic cod to recover from low stock sizes despite the moratorium on fishing can be attributed to a range of sources. These include a depensatory effect on recruitment (Rose 2004), combined with climate change and associated ecosystem changes (Rice 2002), increased predation (Fu *et al.* 2001), and the low-level bycatch of the species under conditions of low productivity (Shelton *et al.* 2006). The range of hypotheses suggests that no single factor can be identified, and the range of potential factors may continue to prevent recovery.

Sablefish

Sablefish *Anoplopoma fimbria* is only found in the northern Pacific, with greatest concentrations located in the Gulf of

Alaska and Bering Sea (the northern stock, the focus of this case study). Demersal adults inhabit depths >400 m, and prey on small fish, squid, crabs and worms. The species is long-lived (>70 years), although most fish are mature by six years old. Recruitment to the population varies from year to year, probably related to varying supplies of copepod prey for the juveniles (McFarlane and Beamish 1992) and to variation in water temperatures (Sigler *et al.* 2001). These fluctuations have important impacts on the fishery more than four years later.

Following heavy exploitation of groundfish by foreign vessels and their subsequent exclusion from Canadian and USA Exclusive Economic Zones (EEZs), domestic fishing effort on sablefish increased dramatically. Catches are mainly by longliners and trawlers, and traps (set on longlines) are also important in Canadian waters. In the 1980s, catches increased substantially as a consequence of both increasing fishing effort and strong incoming year classes. Population levels subsequently declined, however, as a result of the large amount of fishing effort in the open-access fishery. To reduce catches, the fishing season was shortened, but continuing declines in population levels led to further restrictions in season length. The Alaskan season was reduced from eight months in 1984 to nine days in 1994, while the British Columbia fishing season ended up as a 14-day fishery, in other words 'Olympic' sprints to catch as many fish as possible before the next boat. This often led to the quota being exceeded, and also affected market prices owing to the short-term oversupply of fresh fish, much of which was subsequently frozen and sold as lower-value product.

In an attempt to solve this problem, Canadian authorities introduced a limited-access fishery, governed by individual vessel quotas (IVQs), in 1990. Alaskan authorities took the same approach in 1995, with the introduction of 'individual fishing quotas' (IFQs) (Sigler and Lunsford 2001). Although still closely controlled by Government, the approach allowed fishers to use their quotas to fish when they wished to, allowing more constant market conditions and a steadier supply of higher value product. The process also increased the level of interaction and consultation between fishers and managers, and both USA and Canadian assessment systems include sablefish fisher representatives within the process.

Catch data are considered to be fairly accurate in both Canadian and USA fisheries, with data collected using logbooks and observers on both sablefish-targeting vessels and those catching the species as bycatch. Moreover, independent dockside monitors in British Columbia observe 100% of the landings, and vessels are required to carry VMSs. Abundance and biology are also assessed through regular standardised fishery-independent surveys, and supporting holistic studies have examined the impact of the environment on recruitment patterns (e.g. Schirripa and Colbert 2006). TACs are based on the results of regular stock assessments and stock projections, and set at the maximum permissible yield that will maintain agreed levels of spawning-stock biomass with high probability. Assessment decision tables may be explicitly displayed in terms of the trade-offs of potential management decisions in terms of

conservation, stability and yield (e.g. Haist *et al.* 2005) when considering future developments within the fishery. Each quota within the TAC defines a guaranteed share. When the TAC is reached, the fishery is closed, demonstrating swift management response time. The management decision-making process has implicit if not explicit procedures, and is largely defined by national regulations and frameworks (e.g. the Magnuson-Stevens Act in the USA).

The increased catch rates, decreased harvest of immature fish and improved catching efficiency have been attributed to the move to IFQs, with the longer fishing season that results leading to reduced competition between fishers and increased incentives for long-term investment and efficiency (Hilborn *et al.* 2005). In turn, the collaborative approach between scientists and fishers has facilitated joint research, stock assessment, management and enforcement, with funding for these activities derived from licence revenue, thereby improving the information available for assessment and management.

South African West Coast rock lobster

The South African fishery for West Coast rock lobster *Jasus lalandii* started in the late 19th century, although archaeological finds of middens show that the species was already being taken 10 000 years ago (Buchanan 1988). Commercial catches gradually rose throughout the early 20th century to some 9 000–11 000 t annually by the 1950s, then decreased stepwise as stocks became less productive and a degree of overexploitation took place (Pollock 1995). The stock declined despite quotas and minimum carapace-length limits being introduced progressively (Cockcroft and Payne 1999). Resource productivity was strongly affected during the 1990s when somatic growth rates decreased, and large mortalities (up to 2 000 t at a time) were recorded as lobsters simply ‘walked out’ of the sea, seemingly to escape anaerobic conditions offshore generated by blooms of phytoplankton (Cockcroft 2001). The recent declines took place at a time when democratisation of the country demanded that a greater share of the resource be made available to formerly disadvantaged fishers and fishing communities, at the expense of the large fishing enterprises that had benefited from the resource throughout the 20th century (Cockcroft and Payne 1999).

Clearly, West Coast rock lobster management needed to be dynamic and able to respond to fishery and external drivers, as well as to policy, if it was to remain sustainable. Management and policy have seemingly been quite successful, employing just about all the tools listed by Garcia (1994). Both precautionary management and clear use of the best scientific evidence on the stock certainly apply to current management policy, as does the adoption of a risk-averse strategy in setting TACs and other management measures and controls (Cockcroft and Payne 1999 and references therein). Managers also clearly agree on up-front acceptable levels of impact and risk, as well as supporting holistic management of the resource in its known nearshore environment (Shannon *et al.* 2006). Where necessary too, management reacts swiftly in addressing issues and potential problems, and carries out an immense

amount of prior consultation across the board. MCS (e.g. observers, port sampling, coastline and factory inspections) of this highly accessible resource is in fairly good shape, despite the recent activities of poachers in part of its distributional range, which is highly publicised in the local media. Perhaps less successful, however, has been the proven adoption of the sustainable development principle for the stock, possibly because the stock was recovering from a nadir just 10 years ago anyway, nor too have clear criteria been set to assess the impact of development. Further, despite operational decision-making procedures and the very recent agreement and establishment of an operational management procedure for the stock (Johnston and Butterworth 2005), one cannot with conviction say that a range of management benchmarks and reference points has been considered other than by scientists, who have been driving the process. Non-fishery users have also not truly advanced in South Africa far beyond their role as lobbyists, so their participation in management *per se* is still not as great as many would wish.

The fact that the resource has remained in reasonable shape, despite the political pressures for increased allocations to needy fishers and fishing communities while keeping a good portion of the pre-democracy fishing enterprises viable, does show that fishery management can work if the available tools are utilised well and if there is buy-in of the majority of participants to the objective of sustainability. Indeed, the TAC has risen slightly since the turn of the millennium. Cockcroft and Payne (1999) concluded that management of West Coast rock lobster in South Africa was dynamic and ‘striving to conform to precautionary management principles’. Of course, the TAC (now just over 2 200 t) is still well below the ~10 000 t annual yield up to the mid-1960s, but declines have been halted and sustainability seemingly achieved, despite the potential for increased fishing effort.

South Africa competes with no other country for its West Coast rock lobster resource, neighbouring Namibia managing its own sector of the resource. This is a huge advantage over the situation for some of the other case studies outlined here. However, perhaps the greatest achievement of fisheries management in South Africa, exemplified by the apparently sustainable management of West Coast rock lobster, is the scientific, administrative and political will to develop and employ operational management procedures (OMPs), developed and understood by most and transparent to all (Geromont *et al.* 1999, Johnston and Butterworth 2005).

Performance of the four stocks in terms of precautionary management

For each of the four case study fisheries, we have summarised in Table 1 our personal opinion of their performance against the 13 precautionary management tools listed by Garcia (1994). The criteria used to reach our summary opinion for each stock are, of course, largely subjective, based on our personal knowledge of the four case study stocks and perusal of others’ views in the literature. Our ‘evidence’ is given concisely in the four case study summaries above,

so we do not repeat it here; Table 1 simply summarises our opinion. However, subjectivity is the main reason we use just three categories of presumed adherence to Garcia's stated principles of precautionary management: 'probably not', i.e. little evidence that we could find of it being invoked; 'possibly', namely that there is some evidence in the literature and the management performance; and 'almost certainly', i.e. that there is ample evidence in the literature and the performance of management. We believe that other scientists and managers would reach similar conclusions based on the information we had before us.

What is clear from our simplistic summary conclusions is that, based upon the small cross-section of fisheries examined, the greater the number of ticks (ticks meaning proven performance) assigned to the 13 tools, the better the overall stock status seems to be in terms of historical performance (the bottom row of Table 1), i.e. more ticks equals better current status. However, scoring highly under one or two particular tools does not necessarily lead to stock health and sustainable exploitation; some of the tools appear to be more important than others in enhancing sustainability through management. In particular, management needs to be risk-averse for exploitation to be sustainable (some may say that this is obvious), but whereas best scientific evidence may be needed to demonstrate overexploitation, rigorous MCS alone cannot solve the problem of declining stocks without a sustainable and precautionary management target in place to guide its activities.

Discussion

The components required within fisheries management to achieve sustainability are changing. Historically, management has been top-down, the main driver being the economic viability of a fishery. Management was supported by the science it saw as necessary to fulfil that driver.

Fishery management failures, and a maturing of scientific knowledge, have changed that view. Indeed, it should be noted that collapse of the northern Atlantic cod stock took place before the effective introduction of the precautionary approach by the FAO. The collapse of that stock and the lessons learned added weight to the precautionary approach and led to new developments in fisheries science (Caddy and Agnew 2004). In turn, acknowledgement of the conflict between the plethora of fisheries drivers (biological, economic, social and political; Grafton *et al.* 2007, Hilborn 2007) has helped to raise serious issues that need to be addressed in developing management approaches. Change in fisheries management is accelerating, however, with the development of new paradigms and drivers that threaten to confuse further the underlying issues.

The World Summit on Sustainable Development in Johannesburg (COFI 2003) called for stocks to be maintained or restored 'to levels that can produce the maximum sustainable yield ... where possible not later than 2015'. Whether the principle of maximum sustainable yield (MSY) actually equates to sustainable management in a fluctuating environment is a discussion that has been continuing for many years (Larkin 1977, Walters *et al.* 2005). The Ecosystem Approach to Fisheries (FAO 2003) and the management of illegal, unreported and unregulated fishing (Kirkwood and Agnew 2004) represent further disparate developing drivers for management. To achieve a sustainable future for fisheries, all approaches will require better use of existing information and/or the development of new information, and their incorporation within existing or new assessment and management systems.

Future fisheries managers and advisers must juggle these and other conflicting drivers in an appropriate manner to ensure stock and ecosystem sustainability. However, the question remains: how can sustainability be achieved within the changing landscape of modern fisheries management?

Table 1: Summary of authors' opinion as to whether the resources covered by the four case studies discussed here have utilised the precautionary tools for ensuring sustainable fisheries (Garcia 1994) during the past few years

Precautionary tool	North Sea cod	Northern Atlantic cod	Sablefish	South African West Coast rock lobster
Adopt the sustainable development principle	x	x	✓	✓
Adopt precautionary management	✓	✓	✓	✓✓
Use best scientific evidence	✓✓	✓✓	✓✓	✓✓
Adopt range of management benchmarks and reference points	✓	✓✓	✓✓	✓
Criteria to assess impact of development	✓	✓	✓	✓
Risk-aversion	x	x	✓	✓✓
Agree acceptable levels of impact and risk	✓	x	✓	✓✓
Accept holistic view of resources in environment	✓✓	✓	✓	✓✓
Swift management response time	✓	✓	✓✓	✓✓
'Non-fishery users' in management bodies	✓	x	✓	✓
Improve decision-making procedures	x	✓	✓	✓
Introduce prior consultation procedures	✓	x	✓✓	✓✓
Strengthen MCS	✓✓	✓✓	✓✓	✓✓
Overall status in terms of sustainability of resource	x	x	✓	✓✓

x Probably not

✓ Possibly

✓✓ Almost certainly

As our short review seems to show, there are a number of key components of successful management. These focus, *inter alia*, on the inclusion of stakeholders within the management process, the need for holistic data collection, and the nature of management (Figure 2), and we discuss these three further below.

Stakeholder involvement

From the material examined here and elsewhere in the literature, a common factor in success stories is the involvement and cooperation of all stakeholders (fishery managers, scientists, fishers, non-governmental organisations, etc.) within the management process. When a fishery is under pressure and declining, so that sustainability at even depressed levels of resource use is jeopardised, adversarial stances tend to be taken by practitioners of science on the one side, and active fishers, their communities and the fishing industry on the other. This is not good for optimal management of resources, let alone sustainability, and decision-makers then have a difficult job evaluating the demands of both sides, and mistakes can easily be made (Daw and Gray 2005). Consequently, the establishment of effective partnerships between fishers and scientists is a healthy development that raises the credibility of management for all players.

The idea of partnerships between science and industry probably started in Peru at least 40 years ago, but those endeavours (known as *Eureka* surveys) were as much about maximising the economic benefit to industry as about optimally managing for sustainability. Present-day partnerships

between science and industry, however, invariably try to address failings in data collection, to seek new technology or potential new resources to harvest, and to provide fresh data that sometimes add weight to long-term trends, and sometimes show the exact opposite! Good examples of effective partnerships (working together for common benefit and fisheries sustainability) exist in Australia (Smith *et al.* 1999), New Zealand (Harte 2001), the USA (Hartley and Robertson 2006), Canada (Zwanenburg and Wilson 2000), the Netherlands (Johnson and van Densen 2007) and the United Kingdom (Armstrong *et al.* 2008). The last authors summarise the partnership phenomenon worldwide, and list the drivers behind the establishment of such partnerships, commenting on much that is good about them, but also stressing their limitations. They conclude that formal programmes of collaborative research are important and successful at bringing fishers into the scientific research (and management) process, and that together with broad-based involvement of fishers in science-led cooperative research, the programmes contribute substantially to the successful design and implementation of conservation measures that have credibility with all stakeholders as well as the public. Therefore, if correctly utilised, such partnerships can underpin the sustainability aim of fisheries management.

Best scientific information

Best scientific information is not necessarily required for sustainable exploitation; precautionary management can lead to sustainable exploitation with minimal information, although

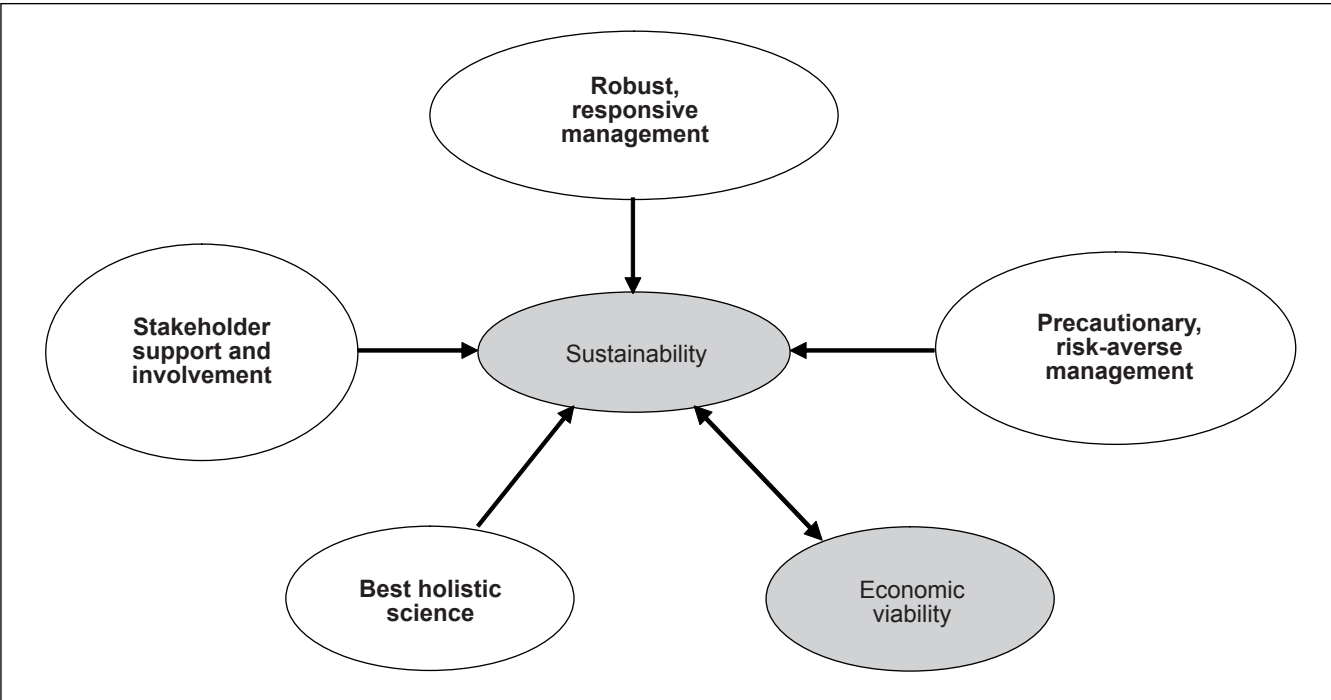


Figure 2: Components of sustainable fisheries. The four emboldened components are felt by the authors to be the most important in achieving sustainability and its necessary corollary, economic viability (shaded components)

yields are likely to be small as a result. Further, though, the question has been raised as to whether sustainability is indeed even achievable for fisheries management, no matter how good the science (Holt 2006). Notwithstanding this point, most would attest that the greater the knowledge and the less the uncertainty, the greater *may* be the yield obtained from a resource or a system. Certainly in the case of northern Atlantic cod, a better understanding of the ecosystem interactions and patterns of fishing may have improved the evidence on which management decisions were based. The partnership process mentioned above enhances this search for knowledge. However, we stress the point that knowledge alone will not generate sustainable exploitation, if indeed it is possible — managers have to act upon it. Also, as scientists we should at least acknowledge that best scientific information can sometimes lead us to advise wrongly or misguidedly. The northern Atlantic cod stock is an example of scientific advice being perhaps erroneous, even if the stock's performance was being confounded by other drivers than fishing alone (Shelton 2005).

Management

To achieve sustainability, managers and stakeholders need to be willing and able to act definitively where best scientific evidence indicates that action is necessary. This will require mutually agreed fishery goals and acceptable levels of risk to be defined clearly. To this end, structured stakeholder workshops in Western Australia have been used to assess the risks involved within particular issues and to identify and agree whether explicit management actions or modifications of existing arrangements are necessary. This approach significantly improved stakeholder involvement and therefore acceptance of outcomes (Fletcher 2005). The same principle applies in the collaborative development and implementation of OMPs, such as applied in South Africa and elsewhere (Geromont *et al.* 1999, Payne and Bannister 2003, Johnston and Butterworth 2005).

Once acceptable levels of relative risk have been defined and agreed among stakeholders, an operational framework needs to be put in place to ensure that potentially harmful levels of exploitation are avoided given uncertainties within the system. Actions to be taken for a given situation, within a fishing season if necessary, need to be pre-agreed among all stakeholders, and these agreed actions need to be combined with robust management approaches, operationalised within management plans or OMPs that can be tested through management strategy evaluation simulation (Kell *et al.* 2005a, De Oliveira *et al.* 2008) against existing uncertainties.

There is no magic bullet, however. Specific fisheries require specific approaches to solve the issues inherent in the resources they exploit. That is very clear from the case studies presented here. Some fisheries are doing better than others, though, and beyond the key attributes already noted, it is not necessarily immediately obvious as to why. As an example, ITQs, currently used successfully in the sablefish fishery, may lead to conflict in fisheries where the rights to fish are hampered by alternative uses of the sea, such as

protected areas or aquaculture (Gibbs 2007). There will be benefits and costs involved with all potential management approaches, and they vary according to the weight assigned to each precautionary tool in each approach. Therefore, it is up to all stakeholders, not just the managers, to agree which approach results in the best balance between favourable outcomes and the minimum of drawbacks for their fishery and ecosystem, under the precautionary approach and the tools detailed by Garcia (1994).

Fishery management inevitably becomes more complicated when it is on a regional scale, but evidence suggests that management also becomes less effective. The South African West Coast rock lobster, along with fisheries in Iceland, Australia and New Zealand, which could also be expressed as 'successful', have in common the fact that the fisheries are managed by a single country, though they may be prosecuted by various (legal and illegal; commercial and recreational) fishing entities. Although our case study sablefish stock is managed by the USA and Canada, strong binding agreements exist between the two countries. North Sea cod, however, are managed under the Common Fisheries Policy (CFP) of the Commission of European Communities. Within this process, individual countries have the ability to maximise the benefit to their own fishers and to play one exploited stock off against another in an attempt to gain greatest benefit from key resources. Management is weakened and, as in the case of North Sea cod, the ability to achieve sustainable resource exploitation may be jeopardised.

During the 2002 reappraisal of the CFP, an attempt was made to improve the system of governance by increasing the amount of stakeholder participation, decentralisation, transparency, accountability, effectiveness and coherence. However, Gray and Hatchard (2003) concluded that this attempt to improve the quality of governance was 'more rhetorical than real'. As a result, sustainability and fisheries management may be irreconcilable where regional management holds sway, but may both be achievable where strong and binding agreements exist between few countries, or where one country has total control over its own fishery and hence the destiny of that fishery.

Ultimately, precautionary, robust and responsive management is needed, developed in agreement with stakeholders, based on adequate scientific information, to achieve sustainability and economic benefit (Figure 2). What appears clear, however, is that, under current conditions, the definition of sustainability is likely to equate to lower, though potentially more profitable, catches than in the past (Hilborn 2007), if, as Holt (2006) questions, sustainability is a notion rather than a myth.

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